

# Work Experience Document

Final Interview Submission: Senior Manager – IT

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**Executive Overview:** A comprehensive technical experience portfolio mapping over 20 years of technology leadership, systems architecture, mission-critical infrastructure, and IT governance directly to the operational mandates of Obayashi Corporation.

## Enterprise Systems & High Availability

Proven track record engineering safety-critical platforms (Philips Healthcare / Cyient) with 99.95%+ uptime, automated failover, and zero-loss offline data resilience (Technoyana).

## Cybersecurity, Risk & Compliance

Deep expertise in Zero-Trust Network Access (ZTNA), RBAC, ISO 27001 standards, vulnerability remediation, and continuous disaster recovery readiness.

## Project Execution & Digital Innovation

Extensive experience modernizing legacy systems, managing agile cross-functional squads, and developing spatial 3D/BIM-aligned technologies.

## Vendor Governance & Fiscal Discipline

Demonstrated capability managing multi-vendor contracts, cloud infrastructures (AWS/Azure), SLAs, and driving significant recurring CapEx/OpEx cost reductions.

## 1. Project Overview & Objectives

### Project 1: Mission-Critical Clinical Systems (Philips Healthcare / Cyient)

**Role & Tenure:** Senior Technical Lead / Systems Architect (Nov 2023 – Mar 2025)

**Scope & Scale:** Philips Intellispace Critical Care and Anesthesia (ICCA)—an enterprise-grade healthcare monitoring and clinical decision support system deployed across major hospital networks globally.

**Core Objectives:**

- **Zero-Downtime Reliability:** Guarantee fail-safe system availability and instantaneous clinical alert rendering in 24/7 ICU environments where system latency carries life-safety consequences.
- **Security & Regulatory Compliance:** Implement strict healthcare data governance (HIPAA, FDA software safety benchmarks, ISO 27001 data classification, and end-to-end cryptographic transit).
- **Modular Architecture Modernization:** Transition legacy monolithic modules into scalable micro-frontends and standardized API communication layers, accelerating release cadence without regression risk.

### Project 2: High-Availability Cloud Modernization & Edge-Resilient Platform (Technoyana Digital)

**Role & Tenure:** Co-Founder & Director of Technology / Architecture Lead (May 2021 – Oct 2023; Strategic Advisor)

**Scope & Scale:** Distributed multi-tenant platforms (Twitan Sports OS, automated bracket engines, and spatial AI pipelines) operating across physical venues with degraded cellular networks.

**Core Objectives:**

- **Offline-First Field Resilience:** Engineer a zero-loss offline architecture using local client persistence and Service Workers that allows uninterrupted field operations during network outages, synchronizing automatically when reconnected (*directly applicable to Obayashi's remote construction sites*).
- **Cloud Cost & Infrastructure Governance:** Directed cloud infrastructure roadmaps across AWS, Firebase, and serverless architectures, automating scaling and reducing recurring cloud expenditures by 22%.
- **Spatial & 3D Engineering (Srushtilabs):** Researched and deployed automated 3D mesh workflows and WebGL visualization pipelines—establishing direct technical continuity with Obayashi's Building Information Modeling (BIM) and Digital Twin roadmaps.

## 2. Role & Core Responsibilities

### IT Strategy, Policies & Architecture

- Formulated multi-year enterprise technology roadmaps aligning IT investments with corporate growth.
- Established standardized IT governance policies, coding standards, and change management protocols.
- Conducted Architecture Review Boards (ARBs) to eliminate technical debt and prevent vendor lock-in.

### Infrastructure Operations & Business Continuity

- Ensured 24/7 availability of critical multi-tier infrastructure, cloud tenancies, and network topologies.
- Supervised automated multi-region backup regimens and formal Disaster Recovery (DR) tabletop drills.
- Implemented proactive synthetic monitoring and telemetry to preemptively resolve system anomalies.

### Cybersecurity, Access Control & Risk

- Enforced Zero-Trust principles, role-based access control (RBAC), and mandatory multi-factor authentication (MFA).
- Executed vulnerability scans (SAST/DAST), code audits, and third-party vendor risk assessments.
- Maintained disaster containment playbooks and compliance with ISO 27001 and industry safety standards.

### Team Leadership, Vendors & Budgets

- Led, mentored, and developed cross-functional squads across development, infrastructure, and QA.
- Governed multi-vendor contracts, CSP relationships, and enforced strict SLA compliance.
- Optimized CapEx/OpEx allocations through license rationalization and cloud cost management.

### 3. Major Achievements & Measurable Milestones

| Milestone / Metric           | Result     | Operational Context & Measurable Impact                                                                                            |
|------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------|
| Critical System Availability | 99.95%+    | Maintained 24/7 continuous uptime across clinical ICU operations and high-traffic event workloads without single-point failures.   |
| Edge Data Preservation       | 100%       | Zero data loss recorded during remote venue network dropouts using offline-first state engine and transactional event queueing.    |
| Release Velocity             | +35%       | Accelerated deployment and release cycles from weeks to hours via automated CI/CD pipelines, containerization, and linting suites. |
| Cloud Cost Optimization      | 22% Saved  | Reduced recurring monthly cloud spend through proactive rightsizing, autoscaling policies, idle asset pruning, and license audits. |
| Incident MTTR                | 40% Faster | Cut Mean-Time-To-Resolution (MTTR) by implementing proactive synthetic monitoring and standardized incident escalation runbooks.   |
| Security Audit Cleanliness   | Zero Highs | Successfully cleared rigorous enterprise and healthcare security audits with zero unresolved high-severity vulnerabilities.        |

### 4. Challenges Faced & Resolutions Implemented

#### Challenge 1: Severe Network Fluctuation & Remote Field Disconnections

**Context:** Remote venues and distributed client sites frequently suffered from unstable cellular coverage, local Wi-Fi interference, and sudden ISP dropouts, risking broken transactions and user data loss.

**Resolution:** Re-architected data ingestion to an **Offline-First model** using client-side IndexedDB persistence and Service Workers. Implemented an optimistic UI pattern with an automated transactional replay queue. When offline, transactions are stored locally with cryptographic timestamps; upon reconnection, the queue replays in the background with deterministic conflict resolution.

**Impact:** 100% data preservation, zero operator workflow interruptions, seamless user experience.

#### Challenge 2: Architectural Fragility & High Downtime Risk in Monolithic Legacy Systems

**Context:** Critical enterprise applications were bound within monolithic codebases where simple updates risked triggering cascading failures across unrelated modules, leading to long testing cycles and deployment anxiety.

**Resolution:** Decomposed the monolith into decoupled **micro-frontends** and standardized REST/WebSocket API contracts. Established automated test pipelines and canary / blue-green deployments with instant rollback capability.

**Impact:** Reduced regression risks to near zero, cut deployment release cycles by 35%, and enabled independent modular upgrades.

### Challenge 3: Complex Multi-Vendor Dependencies & SLA Enforcement Gaps

**Context:** Proliferation of disjointed third-party cloud tools, software licenses, and ISP vendors led to finger-pointing during service degradation, lack of unified visibility, and unchecked budget leakage.

**Resolution:** Designed and deployed an **IT Operational & SLA Governance Dashboard** providing real-time telemetry on vendor uptime, latency, and support turnaround times. Renegotiated MSAs to tie vendor billing directly to SLA compliance scores and automated the termination of idle cloud instances.

**Impact:** Slashed incident MTTR by 40% and trimmed recurring monthly infrastructure expenditures by 22%.

## 5. Lessons Learned & Operational Best Practices

- **Design for Failure & Network Resilience from Day 1:** In mission-critical operations—whether in clinical hospital wards or remote construction job sites—assuming continuous 100% network uptime is an architectural flaw. Systems must inherently feature local caching, asynchronous retry queues, and graceful degradation modes.
- **Cybersecurity Must Be Cultural, Not Merely a Perimeter Firewall:** Modern enterprise defense requires the Zero-Trust paradigm: continuous verification, least-privilege access, automated multi-factor authentication, and regular staff hygiene training. Protecting intellectual property and project blueprints requires data-level encryption and access auditability.
- **Proactive Observability Over Reactive Incident Response:** An IT organization should never learn about a system outage from an end-user ticket. Automated synthetic transactions, threshold-based health alerts, and centralized log telemetry enable IT teams to resolve 80% of system degradation before business operations are affected.
- **Bridging Field Realities with Executive Strategy:** Effective IT leadership is achieved in the field. Understanding the daily friction points of site engineers and project managers ensures technology investments solve tangible business problems and deliver measurable ROI.

## 6. Suggestions for Future Process Improvements (Tailored for Obayashi Corporation)

### 1. Construction Site "IT-in-a-Box" Rapid Deployment Kit

Develop containerized, pre-configured site IT units featuring dual-SIM bonded 5G/satellite SD-WAN, localized edge caching nodes, biometric access control, and ruggedized Wi-Fi 6 mesh. Enables full job-site IT commissioning within 48 hours via Zero-Touch Provisioning (ZTP).

### 2. Cloud BIM & Common Data Environment (CDE) Acceleration

Deploy localized on-site BIM caching nodes paired with WebGL-based progressive 3D rendering pipelines. Allows field engineers to inspect multi-gigabyte 3D models and digital twins on mobile tablets with zero latency, synchronizing annotations back to corporate CDE asynchronously.

### 3. Subcontractor Zero-Trust Security Mesh

Establish a cloud-native Zero-Trust Network Access (ZTNA) portal granting subcontractors strictly time-bound, role-scoped access to authorized project work packages with dynamic watermarking on proprietary blueprints and commercial bids to eliminate data leakage.

### 4. AIOps Predictive Maintenance & Automated DR

Implement AI-assisted log telemetry across regional hubs and active construction sites to predict hardware/network failures before outages occur. Couple with immutable air-gapped backups to guarantee ransomware protection and automated quarterly disaster recovery drills.

## First 90 Days: Strategic Execution Roadmap for Senior Manager – IT

#### Days 1 – 30: Discovery & Baseline

- Audit all IT infrastructure, multi-site networks & cloud tenancies.
- Interview site project managers & business heads for pain points.
- Review cybersecurity posture, MFA enforcement & DR readiness.
- Benchmark IT budget expenditures, vendor contracts & SLAs.

#### Days 31 – 60: Stabilization & Quick Wins

- Remediate high-risk security vulnerabilities; standardize IAM.
- Resolve immediate job-site network connectivity bottlenecks.
- Implement proactive monitoring dashboards for critical systems.
- Conduct full tabletop Disaster Recovery (DR) simulation drill.

#### Days 61 – 90: Modernization & Growth

- Present 2-3 Year IT Strategy aligned with corporate goals.
- Prototype the "Site IT-in-a-Box" rapid deployment kit.
- Institute automated CI/CD and self-service IT ticketing workflows.
- Establish structured upskilling and mentorship tracks for IT team.

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